Safa.P.S.

AI Engineer | LLM Applications, Multilingual Conversational AI & Self-Hosted Inference

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Summary

AI Engineer with hands on experience building, integrating, optimizing, and deploying LLM and Generative AI applications in production environments. Experienced in Python, PyTorch, Hugging Face, LangChain, RAG, AI automation, multimodal AI, computer vision, OCR, and speech systems. Hands on experience with LLM evaluation and optimization using Gemma, Ollama, LoRA/PEFT, quantization, FAISS, Docker, FastAPI, and local/cloud inference environments.

PROFESSIONAL EXPERIENCE

AI Engineer, Bafar Labs — May 2026 – Present | calicut

- **Built a FastAPI service integrating Twilio WhatsApp with self-hosted Gemma 4 26B and Whisper Large-v3-Turbo** to replace Gemini in Ziya Gold's AI sales assistant, handling text, voice notes and photos.
- **Deployed Gemma on a RunPod RTX A5000 GPU via Ollama**, achieving **~5 s median replies** across **27 live WhatsApp tests**, 10–50× faster than CPU inference.
- Reused the production **prompt, tool calling and 16-message memory** with **multilingual support (English, Arabic, Malayalam)** and lead-tag extraction.
- Added **webhook signature verification, a link-validation guard, dry-run mode and per-message metrics logging** for safe live testing.
- **Evaluated LLM quality on 31 test cases** (Gemma 4 26B: **22/31** against 11/31 for a smaller model) and documented accuracy gaps with fixes.
- Delivered a **cost and scalability analysis of Gemini against a self-hosted GPU**, finding a **break-even at ~51–60k messages a month** and a prompt-caching optimisation.

Add to Technical Skills:

- Built AI-powered automation workflows using **n8n, Gemini, Gmail API, and Google Sheets** for company research, personalized email generation, and automated follow-ups.
- Designed **Day-3 and Day-5 follow-up workflows** with automatic stopping when a prospect replies or an email bounces.
- Integrated **self-hosted Chatwoot using Docker and Cloudflare Tunnel** to manage two-way email conversations within a single CRM thread.
- Engineered and debugged a **full-stack AI video interview platform** using **Next.js, NestJS, Python, LiveKit, and OpenAI Realtime API**.
- Resolved a critical production issue that prevented the **AI interview agent from starting**, restoring the end-to-end interview workflow.
- Worked on **AI image-generation systems**, including prompt engineering, product-fidelity improvements, testing, and backend/frontend debugging.

AI Developer Intern, Banao Technologies — Nov 2025 – May 2026 | Bangalore

- Built an Arabic Text-to-Speech pipeline using Python, Gemini, and gTTS, extending the application to Arabic-speaking users
- Developed Speech-to-Text and Text-to-Speech features using AI APIs for voice-enabled workflows
- Contributed to Interview God, an AI-powered interview assessment platform
- Implemented LLM fallback handling so the system stayed available when primary AI services failed
- Integrated and tested APIs to keep application-level AI workflows stable

AI/ML Engineer Intern, Gwing Software Technologies Pvt Ltd — Oct 2025 – Nov 2025 | Hyderabad

- Developed a multimodal AI chatbot supporting both text and image inputs
- Implemented **LangChain and LlamaIndex** for context-aware conversational AI
- Integrated **Qwen2-0.5B-Instruct and BLIP models** for enhanced NLP and image understanding
- Built local inference APIs using Flask & Streamlit for **low-latency, offline performance**
- Collaborated with cross-functional teams to design and deploy AI features based on business needs

Technical Skills

- **Programming & APIs** — Python, SQL, REST APIs, FastAPI, Flask
- **AI/ML Frameworks:** — PyTorch | TensorFlow | Keras | Scikit-learn | Transformers
- **LLMs & GenAI** — OpenAI | Gemini | Qwen | Gemma | Whisper | Ollama | Prompt Engineering
- **RAG & Tools** — LangChain | LangGraph | LlamaIndex | FAISS
- **Model Optimization** — LoRA, PEFT | Quantization (BitsAndBytes)
- **Multimodal AI & Computer Vision** — OCR | BLIP | YOLO | OpenCV | Vision-Language Models
- **Deployment & Infrastructure** — Docker | RunPod (GPU cloud) | Twilio WhatsApp API | n8n | FastAPI | Flask | Streamlit | MLflow

Projects

RAG-Based AI Chatbot for Educational Institution

- Built a scalable chatbot using **LangChain and OpenAI/Gemini APIs**
- Achieved **95% semantic retrieval accuracy** across 5,000+ FAQs
- Reduced manual support workload by **60%** using FAISS vector search

Self-Healing Sentiment Classifier (Fine-Tuned DistilBERT LLM with LoRA)

- Fine-tuned DistilBERT using LoRA for efficient sentiment analysis
- Implemented fallback logic using LangGraph to ensure consistent output reliability
- Optimized for CPU-only inference with improved efficiency

Arabic Document OCR using Vision-Language Model

- Fine-tuned Qwen2.5-VL-7B-Instruct for OCR tasks using LoRA
- Applied 4-bit quantization and gradient checkpointing for resource optimization
- Built a production-ready OCR pipeline with PyTorch and Transformers

Certification

Data Science & Machine Learning , Illinois Institute of Technology — Jun 2024 – Apr 2025

Education

Bachelor of Science in Information Technology, Mahatma Gandhi University — 2016 – 2019 | Kerala, India